Ummm double slit

1) Where does $n\lambda$ come from?

It doesn't come from the geometry. The geometry just gives you a path difference, which is some continuous number that changes as you move P along the screen. The $n\lambda$ is a condition you impose on top of the geometry. You're saying: "two waves started in phase at the slits. For them to arrive in phase at P and constructively interfere, the difference in distance they travel must be a whole number of wavelengths." That's just the definition of constructive interference, crest lands on crest. So you set path difference = $n\lambda$ and then ask: what does geometry say the path difference actually equals?

Q is where the perpendicular from S_1 meets the S_2 ray. Now, because $D \gg d$, the two rays to P are essentially parallel. That means S_1 and Q are the same perpendicular distance from P. In other words, **$QP \approx S_1P$**. They're on the same wavefront.

2) Why is d the hypotenuse, not the adjacent?

This is where the construction matters. You don't just draw the triangle between the two slits and the screen. You construct it like this: draw rays from S_1 and S_2 to point P. Because $D \gg d$, these rays are nearly parallel. Then from S_1, drop a perpendicular onto the ray from S_2. Call the foot of that perpendicular Q.

Now your triangle is S_1QS_2. The right angle is at Q, which sits somewhere along the ray, not at either slit. Since the right angle is at Q, the side opposite it ($S_1S_2 = d$) is the hypotenuse. The short side opposite θ (from S_1 to Q) is the path difference. So: path difference = $d \sin \theta$.

If you'd accidentally put the right angle at one of the slits, you'd get $d \cos \theta$. The key is that Q is a constructed point on the ray, not a physical object.

3) Why is θ the same in both triangles?

You need θ to appear in two places: the small triangle (giving you $d \sin \theta$) and the big triangle (giving you $\tan \theta = \Delta y/D$). Here's why it's the same angle.

Big triangle: from the midpoint of the slits, draw a line to P. The angle this makes with the horizontal normal is θ, where $\tan \theta = \Delta y / D$.

Small triangle: the ray from S_2 to P makes essentially the same angle θ with the horizontal, because the screen is so far away that shifting from the midpoint to S_2 barely changes the angle. This is the $D \gg d$ approximation.

Now in the small triangle, S_1S_2 is vertical and the normal is horizontal, so they're perpendicular. The ray from S_2 makes angle θ with the horizontal. Since vertical and horizontal are 90° apart, the ray makes angle $(90° - \theta)$ with the vertical slit line S_1S_2. The triangle S_1QS_2 has a right angle at Q, so its three angles must sum to 180°:

$90°$ (at Q) + $(90° - \theta)$ (at S_2) + ? (at S_1) = $180°$

That forces the angle at $S_1 = \theta$. So θ appears naturally inside the small triangle, and it's the same θ as in the big triangle because the parallel ray approximation guarantees the angles match.

The full chain then snaps together: geometry gives path diff = $d \sin \theta$, then impose constructive interference so $d \sin \theta = n\lambda$, then use the small angle approximation $\sin \theta \approx \tan \theta = \Delta y/D$, substitute to get $d \cdot \Delta y/D = n\lambda$, and solve: $\boldsymbol{\Delta y = n\lambda D/d}$.